REPORT 1

Data Preparation

CNN – Medical Image Classification

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1. Introduction

This report documents the data preparation pipeline for the CNN-based medical image classification project. The primary objective of this stage is to load, inspect, preprocess, and augment the Chest X-Ray Pneumonia dataset in preparation for model training. Proper data preparation is a critical prerequisite for achieving reliable model performance, as it directly influences the generalization ability of the trained network.

The steps covered in this report include dataset description, class distribution analysis, train/validation/test splitting, pixel normalization, image resizing, data augmentation, and a methodological justification of each design decision, including a discussion of class imbalance and data leakage prevention.

2. Dataset Description

The dataset used in this project is the Chest X-Ray Images (Pneumonia) dataset, publicly available on Kaggle (Paul Mooney, 2018). It consists of 5,856 anterior-posterior chest X-ray images collected from pediatric patients aged 1 to 5 years at the Guangzhou Women and Children's Medical Center, Guangzhou, China. The dataset is licensed under CC BY 4.0.

2.1 Task Definition The task is a binary image classification problem. Each X-ray image is assigned to one of two classes:

- **NORMAL** Radiologically normal lung appearance.
- **PNEUMONIA** Lungs showing bacterial or viral pneumonia indicators (e.g., consolidation, infiltrates).

2.2 Dataset Statistics

The dataset is pre-divided into three splits as shown in Table 1.

Split	NORMAL	PNEUMONIA	Total	% of Dataset
Train	1,341	3,875	5,216	89.1%
Validation	8	8	16	0.3%
Test	234	390	624	10.7%
TOTAL	1,583	4,273	5,856	100%

Table 1. Class distribution across train/validation/test splits.

2.3 Class Imbalance Analysis

A notable class imbalance exists in the training set: 74.3% of training images belong to PNEUMONIA and only 25.7% to NORMAL. This skew is clinically plausible; medical datasets typically contain more pathological cases, but it poses a risk of bias during training. An unweighted model would likely default to predicting PNEUMONIA for most inputs, yielding artificially high accuracy while failing to correctly identify healthy patients.

To mitigate this, class weights are computed using the balanced weighting strategy (scikit-learn's compute_class_weight). The resulting weights are NORMAL ≈ 1.94 and PNEUMONIA ≈ 0.67. These are applied as sample weights during training, penalizing misclassification in the minority class more heavily.

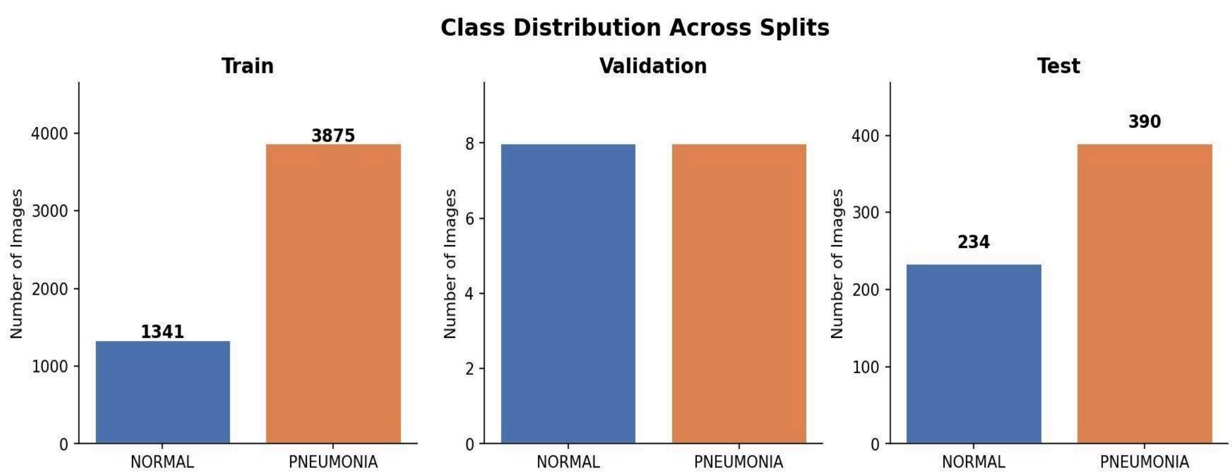

Figure 1. Class distribution across train, validation, and test splits. The training set exhibits significant class imbalance (~74% PNEUMONIA).

3. Data Splitting Scheme

The Kaggle dataset provides a predefined folder structure separating train, validation, and test images. This structure is preserved without modification or reshuffling to ensure experimental reproducibility and prevent data leakage.

Split	Purpose	Augmentation Applied	Shuffle
Train	Model parameter optimisation	Yes, a full pipeline	Yes
Validation	Hyperparameter tuning & monitoring	No rescale only	No
Test	Final unbiased evaluation	No rescale only	No

Table 2. Data splitting strategy and augmentation policy per split.

3.1 Data Leakage Prevention

Data leakage occurs when information from validation or test sets influences the training process, resulting in overly optimistic performance estimates. The following measures are applied to prevent this

- The predefined split is maintained; no images are moved between splits.
- Data augmentation is applied exclusively to the training generator.
- Normalization is applied uniformly using the fixed rescale factor (÷255), not computed statistics. This avoids the need to fit a scaler on training data and reapply it, eliminating a common source of leakage.
- The test set is used only once, at final evaluation.

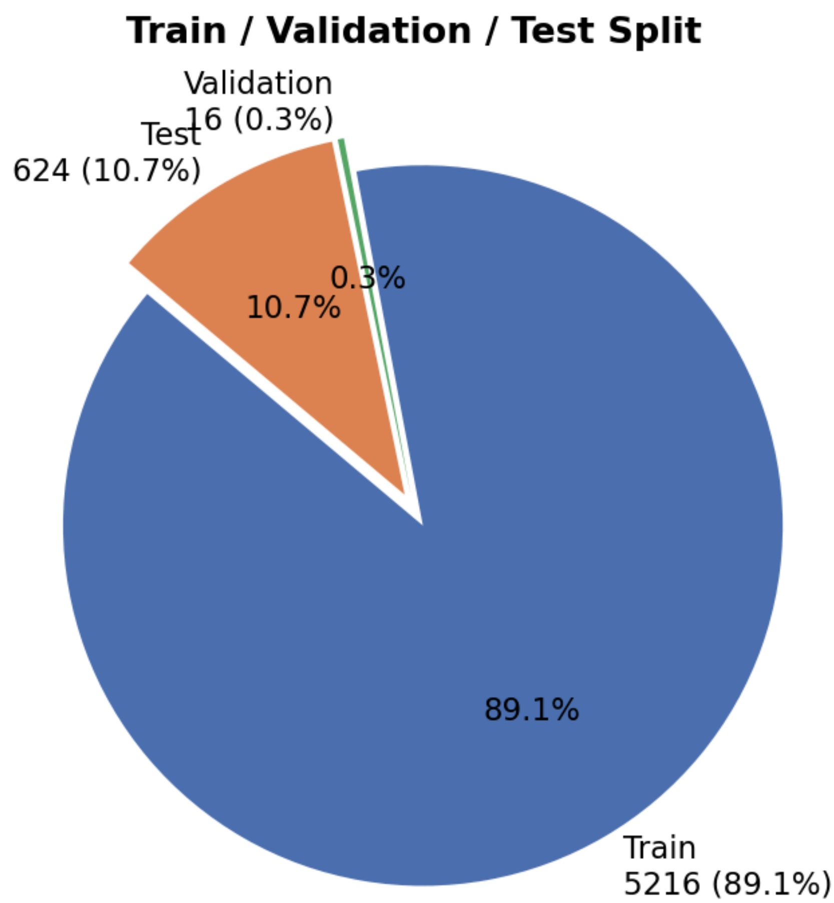

Figure 2. Proportional distribution of the dataset across train/validation/test splits. The training set constitutes 89.1% of all data.

4. Preprocessing Methods

4.1 Image Resizing

All images are resized to 224 × 224 pixels prior to feeding them into the network. This resolution was selected for two reasons: (1) it matches the default input dimensions expected by standard transfer learning architectures such as ResNet50 and VGG16, ensuring compatibility in Report 3; and (2) it provides adequate spatial resolution for identifying radiological patterns associated with pneumonia, such as lobar consolidation and interstitial infiltrates, while remaining computationally tractable on consumer-grade hardware.

4.2 Pixel Normalisation

Raw pixel values in JPEG images range from 0 to 255 (8-bit unsigned integers). These are rescaled to the continuous range [0, 1] by dividing by 255.0. This operation is applied uniformly to all three splits.

$$x\_normalized = x\_original \div 255.0$$

Normalization serves two purposes: it prevents large-magnitude inputs from causing unstable gradients during backpropagation, and it ensures consistent feature scaling across all input channels, a precondition for effective weight initialization strategies such as He or Xavier initialization.

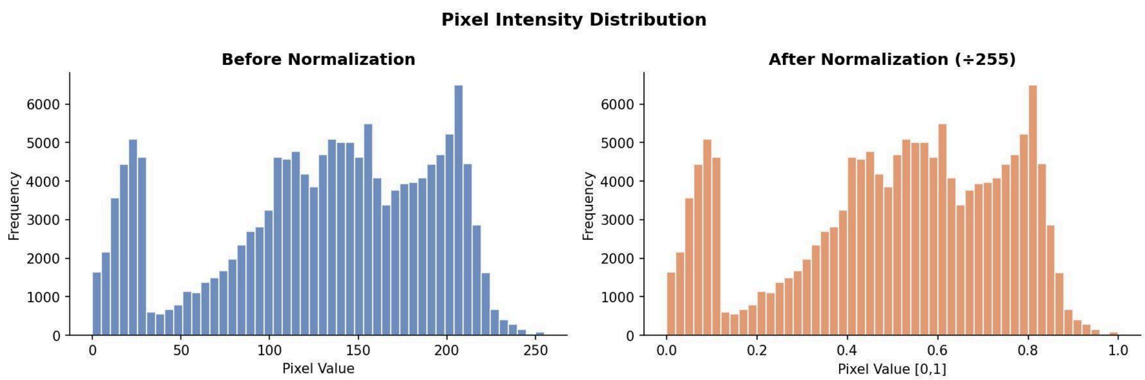

Figure 3. Pixel intensity distribution before (left) and after (right) station normalization. The x-axis range collapses from [0, 255] to [0, 1]; the distribution shape is preserved.

5. Data Augmentation

Data augmentation is a regularization technique that artificially expands the effective training set by applying stochastic geometric and photometric transformations to input images at training time. Each transformation is applied independently with a random parameter drawn from the specified range. Augmentation is implemented using the TensorFlow/Keras ImageDataGenerator and is applied only to the training split.

5.1 Augmentation Parameters and Justification

Transformation	Parameter	Clinical/Technical Justification
Horizontal flip	True	Chest anatomy can appear mirrored depending on patient orientation; both orientations are clinically valid.

Rotation	±15°	Slight rotation simulates variability in patient positioning during X-ray acquisition. Large rotations (>30°) would introduce anatomically implausible orientations.
Width shift	±10% of width	Accounts for horizontal centering variability between imaging sessions and equipment.
Height shift	±10% of height	Accounts for vertical positioning differences.
Zoom	±10%	Simulates variation in the source-to-image distance, which affects apparent organ size in radiographs.
Brightness	[0.8, 1.2]	Simulates differences in X-ray exposure and detector sensitivity across imaging devices.
Fill mode	nearest	Empty border pixels created by geometric transforms are filled with the nearest pixel value, avoiding artificial black borders.

Table 3. Augmentation parameters applied to the training set with clinical and technical justifications

5.2 Excluded Transformations

Vertical flipping and shear transformations were deliberately excluded. An inverted chest X-ray (head pointing downward) does not correspond to any clinically valid imaging scenario and would introduce noise into the training distribution. Shear deformations similarly lack a clinical analogue and could distort anatomical proportions in a way that interferes with the model learning physiologically meaningful features.

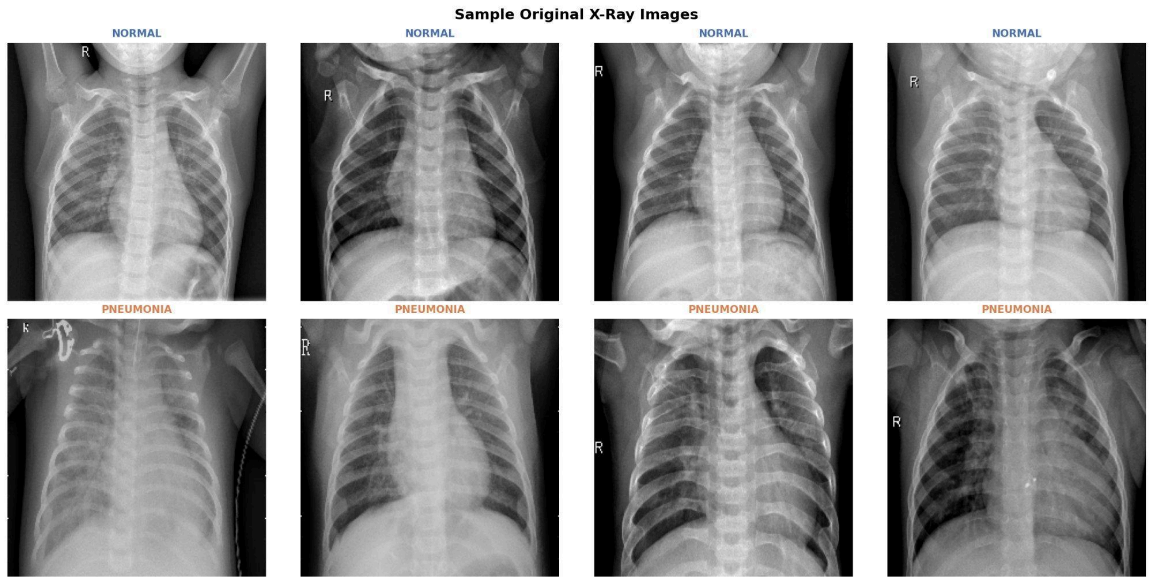

Figure 4. Sample original chest X-ray images from the training set. Top row: NORMAL class. Bottom row: PNEUMONIA class. Pneumonia cases show increased opacity and consolidation.

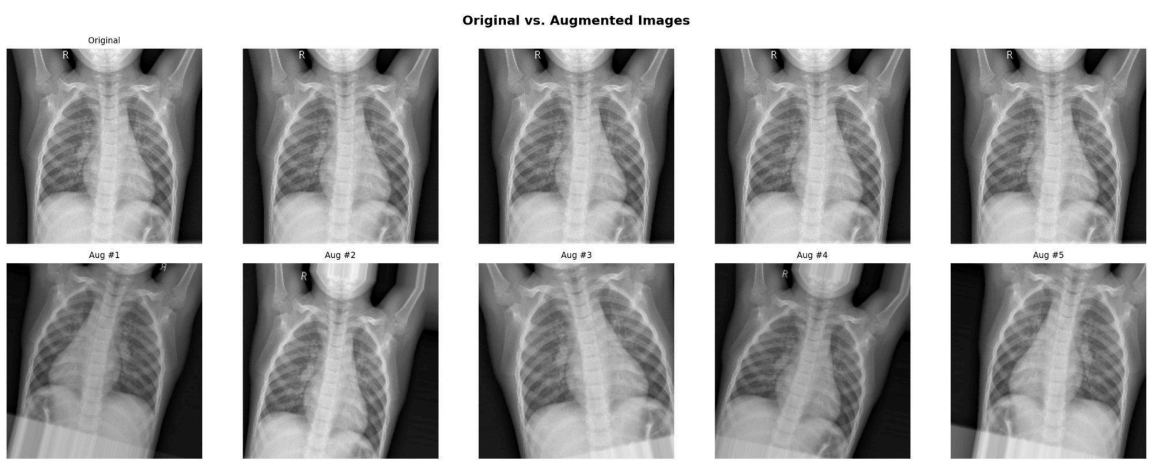

Figure 5. Comparison of the original image (top-left) with five independently augmented variants. Each augmented image applies a random combination of the transformations listed in Table 3.

6. Implementation

The preprocessing pipeline was implemented in Python 3 using the following libraries:

- **TensorFlow/Keras** v2.21.0 model construction, training, and ImageDataGenerator.
- **scikit-learn** v1.8.0: class weight computation and evaluation metrics.

- **matplotlib** v3.10.8 figure generation.
- **seaborn** v0.13.2 statistical visualizations.
- **Pillow** v10.4.0: image loading and resizing.

The complete source code is provided in the file data_preparation.py, submitted alongside this report. All random operations use a fixed seed (seed=42) to ensure reproducibility.

6.1 Generator Configuration Summary

Parameter	Training Generator	Validation/Test Generator
rescale	1/255	1/255
rotation_range	15°	—
width_shift_range	0.1	—
height_shift_range	0.1	—
zoom_range	0.1	—
horizontal_flip	True	False
brightness_range	[0.8, 1.2]	—
target_size	224 × 224	224 × 224
batch_size	32	32
class_mode	binary	binary
shuffle	True	False

Table 4. ImageDataGenerator configuration for training and evaluation splits.

7. Discussion

The data preparation choices described in this report reflect a balance between three competing objectives: maximizing training signal, preventing overfitting, and ensuring clinical validity of the training distribution.

The most significant methodological challenge identified during this stage is the class imbalance in the training set (~74% PNEUMONIA). In a clinical classification context, false negatives (missed pneumonia cases) are typically more harmful than false positives. However, a biased model that systematically predicts PNEUMONIA also endangers healthy patients through unnecessary treatment. Class weighting addresses this limitation by ensuring both error types are penalized proportionally during training, rather than optimizing for aggregate accuracy alone.

The validation set provided by Kaggle contains only 16 images, 8 per class. While this split is maintained for consistency with the dataset's original structure, it should be noted that validation metrics computed on such a small sample will exhibit high variance. Subsequent reports may consider a re-stratified 80/10/10 split from the combined train+val pool if they observe training instability during model development.

8. Conclusions

This report has documented a complete and methodologically rigorous data preparation pipeline for classifying pneumonia in chest X-rays. The key outcomes of this stage are

- 5,856 chest X-ray images organized into train/validation/test splits (89.1% / 0.3% / 10.7%).
- A class imbalance of ~2.9:1 (PNEUMONIA:NORMAL) was identified and addressed via balanced class weights.
- Pixel normalization to [0, 1] and resizing to 224×224 were applied uniformly across all splits.
- A clinically motivated augmentation pipeline (flip, rotation, shift, zoom, and brightness) is applied to training only.
- Data leakage prevention is ensured through strict split isolation and fixed-factor normalization.

The preprocessed generators produced in this stage are directly compatible with the CNN training pipeline and transfer learning models to be developed in Reports 2 and 3.

References

[1] Mooney, P. (2018). Chest X-Ray Images (Pneumonia). Kaggle. https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

[2] Kermany, D. S., et al. (2018). Identifying Medical Diagnoses and Treatable Diseases through Image-Based Deep Learning. Cell, 172(5), 1122–1131.

[3] Chollet, F. (2021). Deep Learning with Python (2nd ed.). Manning Publications.

[4] Shorten, C., & Khoshgoftaar, T. M. (2019). A survey on image data augmentation for deep learning. Journal of Big Data, 6(1), 1–48.

[5] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.